

Frequency response measurement

WAC consortium

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Equations

Definitions:

- Digitizer: measures amplitude $D(f)$, have got frequency dependent gain $G(f)$. Time dependence of the gain is neglected.
- Thermal transfer standard: U is output DC voltage measured by the multimeter, and the overall gain is $\Delta(f)$. Thermal transfer standard and the multimeter are supposed to be stable in time.
- AC source: generates signal of amplitude with non-negligible time and frequency dependence $Z(f, t)$.

One measurement of the AC source by digitizer and by the thermal transfer standard is expressed as:

$$D(f, t) = Z(f, t) \cdot G(f) \quad (1)$$

$$U(f, t) = Z(f, t) \cdot \Delta(f) \quad (2)$$

If the measurement by digitizer and by the thermal transfer standard is done simultaneously, or almost simultaneously, then:

$$D(f) = Z(f) \cdot G(f) \quad (3)$$

$$U(f) = Z(f) \cdot \Delta(f) \quad (4)$$

Two measurements are done, for measured and reference frequencies marked as f_M and f_R . If these measurements are done right after another, then the time dependence of the AC source can be neglected and only frequency dependence is taken into account:

$$t_R \approx t_M \implies Z|_{t_R} = Z|_{t_M} \quad (5)$$

Then we can write both measurements as:

$$D(f_M) = Z(f_M) \cdot G(f_M) \quad (6)$$

$$U(f_M) = Z(f_M) \cdot \Delta(f_M) \quad (7)$$

$$D(f_R) = Z(f_R) \cdot G(f_R) \quad (8)$$

$$U(f_R) = Z(f_R) \cdot \Delta(f_R) \quad (9)$$

Combining the equations:

$$\frac{G(f_M)}{G(f_R)} = \frac{D(f_M)}{D(f_R)} \cdot \frac{U(f_R)}{U(f_M)} \cdot \frac{\Delta(f_M)}{\Delta(f_R)} \quad (10)$$

For f_M varied through whole input bandwidth of the digitizer, the ratio returns the frequency response of the digitizer related to the gain at reference frequency f_R . If the reference frequency is small enough, the gain G_{f_R} can be measured using DC method.

DC offset is not result of measurement and processing!

AC-DC measurements:

$$\Delta_L = 1 + \delta(f) \quad (11)$$

$$U_{ACin} = U_{DCin} \cdot (1 + \delta \cdot 10^{-1}) \quad (12)$$

With:

$$\delta_{792} = \frac{U_{outAC} - U_{outDC}}{n \cdot U_{outDC}} \quad (13)$$

1 Interpolation of the FR function

The FR of the National Instruments 5922 digitizer is of a complex shape. For calculation of SS data, the interpolation of the measured FR data is needed, because the calibration of FR can happen at different frequencies than required for SS measurement.

The FR shape can be interpolated by using polynomial interpolation, however the FR response would require polynomials of a high degree. Such polynomials tends to oscillate between the measured points and especially at leftmost and rightmost parts of the function. The leftmost part of the function is at lower frequency and has the highest impact on the rms value obtained from SS data, thus such interpolation is not suitable.

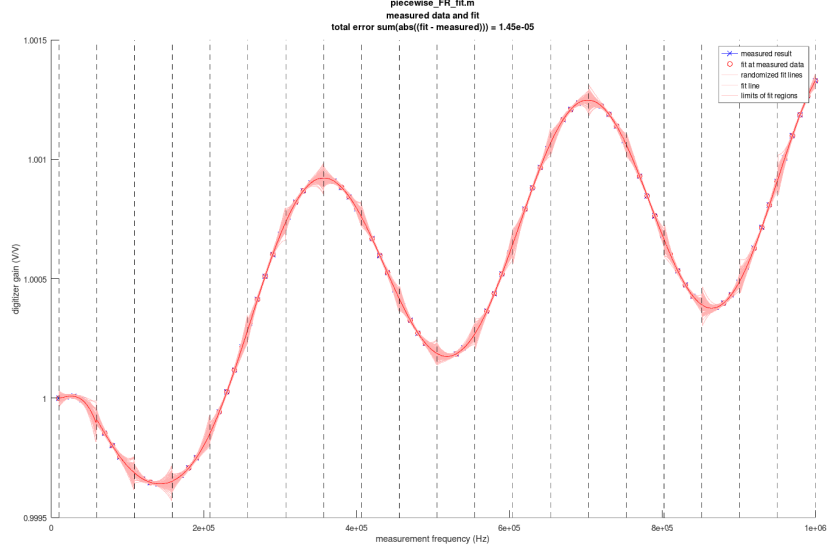


Figure 1: Piece-wise fit of a part of frequency response of a digitizer. The light red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

The possibility is to use piece-wise polynomial interpolation, that mitigates the issue of using high degree polynomials. Naive use of piece-wise polynomial interpolation (just splitting into pieces) does not guarantee smoothness of the whole interpolated function at the edges of the pieces. Later the interpolated curve has to be used also during the Monte Carlo method. Typically when fitting data, the measured points are randomized, multiple fits are estimated, and mean and uncertainty of the fit coefficients are calculated. Later the coefficients are randomized according to the estimated uncertainty, and the uncertainty is propagated through the fit. Randomizing the coefficients of piece-wise polynomials independently cause large discontinuities at the edges of the pieces, see figures 1 2 and 3. To prevent the errors, the randomization of the fit coefficients would need to take into account the requirement of smoothness at the boundaries of the pieces.

GNU Octave function `splinefit` provides simple way to use piece-wise spline (third order polynomials) interpolation, and assures smoothness at the boundaries of the pieces. However still the coefficients cannot be randomized independently. To circumvent the issue, following approach is used. The FR measurement data are randomized and multiple spline piece-wise fits are estimated. Coefficients of the all fits are saved into an intermediate file. The

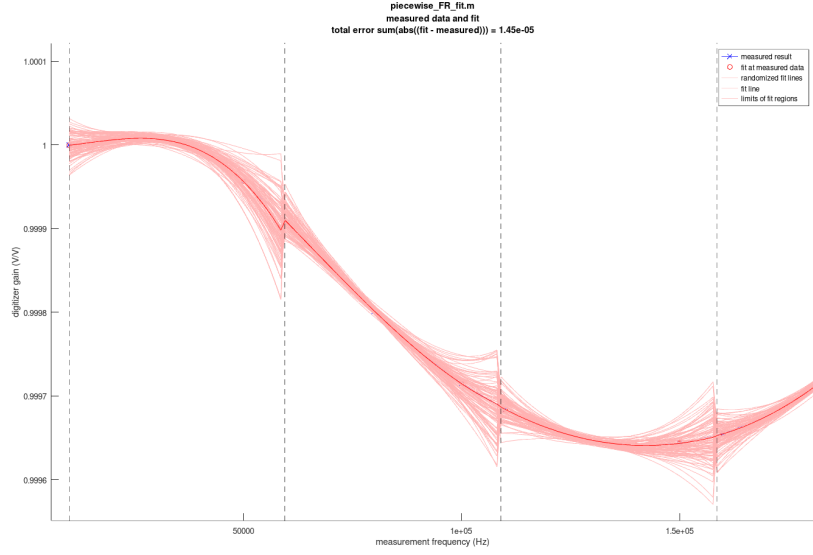


Figure 2: Piece-wise fit of a part of frequency response of a digitizer. The light red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.

intermediate file has to consist of sufficient number of fits to provide good statistics for the uncertainty propagation using Monte Carlo method. Later, when uncertainty has to be propagated through the spline fit, coefficients are loaded and a random selection of the coefficients provides the Monte Carlo method propagation of uncertainties through the fit.

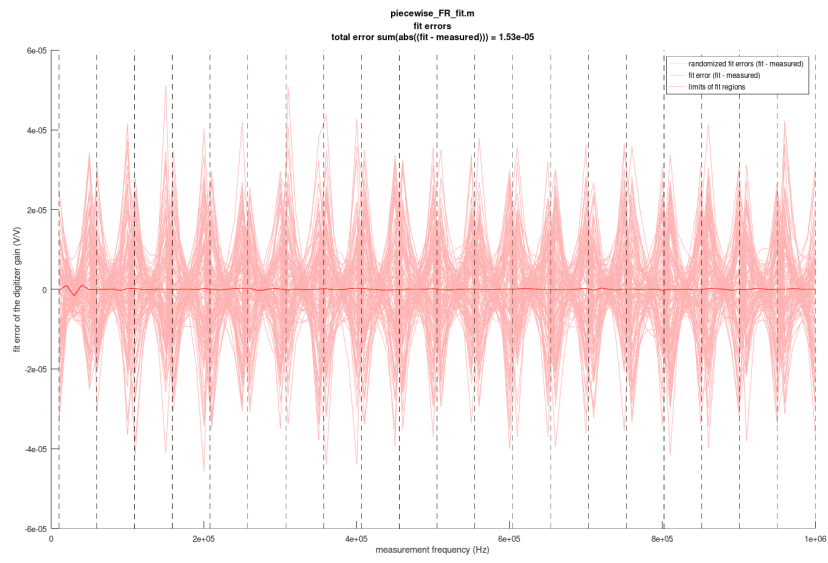


Figure 3: Error of the piece-wise fit compared to the measurement data. The light red represents variations after using Monte Carlo method on the measured data and repeating fit. The vertical lines show boundaries of the pieces.